Optimizing AI Audio Synthesis for Dravidian Languages: A Phonetic Respelling Framework and Strategy for Kannada in NotebookLM

The synthesis of highly localized, colloquially accurate audio content using artificial intelligence presents a significant computational challenge when applied to morphologically complex languages outside the Indo-European family. Standard text-to-speech (TTS) engines, particularly those embedded in platforms like Google NotebookLM, are primarily trained on American English acoustic models and grapheme-to-phoneme (G2P) alignment rules.[1] When these engines encounter transliterated Kannada—a South Dravidian language characterized by agglutination, distinct vowel lengths, and a robust inventory of retroflex consonants—the resulting audio frequently suffers from severe phonetic degradation, misassigned syllable stress, and unnatural prosody.[3]

For the purpose of producing educational podcast content aimed at integrating Hindi-speaking migrants into the Bengaluru sociocultural landscape, phonetic accuracy is not merely an aesthetic preference; it is a pedagogical necessity. A mispronunciation of social honorifics or a failure to articulate the correct imperative verb endings can lead to deep socio-linguistic friction.[5] The objective of this report is to provide an exhaustive framework for overriding standard AI grapheme-to-phoneme interpretation. By establishing a comprehensive phonetic respelling mapping system, American English TTS models can be forced to accurately simulate Bengaluru Kannada phonology. Furthermore, the analysis outlines advanced prompt engineering strategies to manipulate NotebookLM's Audio Overview feature, ensuring the resulting podcast is engaging, culturally nuanced, and structurally optimized for language acquisition.

The Phonological Chasm in AI Speech Synthesis

To effectively correct AI pronunciation, it is essential to understand the specific points of failure where English-trained acoustic models diverge from Dravidian phonology. The Kannada language possesses a phonetic structure that demands specific tongue placements, airflow mechanisms, and durational contrasts that simply do not exist in the American English phonetic inventory.[6]

The Retroflex Consonant Challenge

Kannada utilizes a distinct set of retroflex consonants, which are produced by curling the tip of the tongue backward to strike the hard palate just behind the alveolar ridge.[6] This group

includes sounds like the retroflex 'ṭ' (ಟ), 'ḍ' (ಡ), 'ṇ' (ಣ), and the highly specific retroflex lateral approximant 'ḷ' (ಳ).[8] American English, conversely, relies on alveolar consonants where the tongue tip taps the alveolar ridge directly behind the upper teeth.[7]

When an English TTS model reads transliterated words like *Olagade* (inside) or *Keli* (listen), it defaults to the alveolar /l/, stripping the word of its native Dravidian resonance and producing a sound that is perceptually foreign to native Kannada speakers.[9] While advanced linguistic systems utilize the International Phonetic Alphabet (IPA) to dictate these precise articulations, consumer-facing AI platforms like NotebookLM frequently ignore or misinterpret IPA inputs during audio generation.[12] Therefore, phonetic respelling must creatively manipulate English orthography, vowel weighting, and syllable grouping to artificially induce the heavier, darker resonance characteristic of retroflex articulation.

Contrastive Vowel Length and Aspiration

In Dravidian linguistics, vowel length is strictly semantically contrastive. The duration for which a vowel is held can alter the fundamental meaning of a word; for example, the short vowel in *ka* means 'to learn', whereas the elongated *kaa* translates to 'to protect'.[15] English TTS engines are programmed with stress-timed rhythms, frequently reducing unstressed vowels to a schwa (the neutral "uh" sound), which immediately destroys the acoustic markers of Kannada terminology.[13]

Similarly, Kannada maintains a rigid distinction between unaspirated and aspirated consonants. The unaspirated *ka* (ಕ) requires no extra burst of air, while the aspirated *kha* (ಖ) requires a forceful exhalation.[8] English naturally aspirates voiceless stops only in specific phonetic environments, such as the initial position of a stressed syllable. A standard AI model will struggle to apply aspiration correctly across transliterated Indian texts unless explicitly guided by phonetic respellings that utilize 'h' markers to simulate the burst of air.[15]

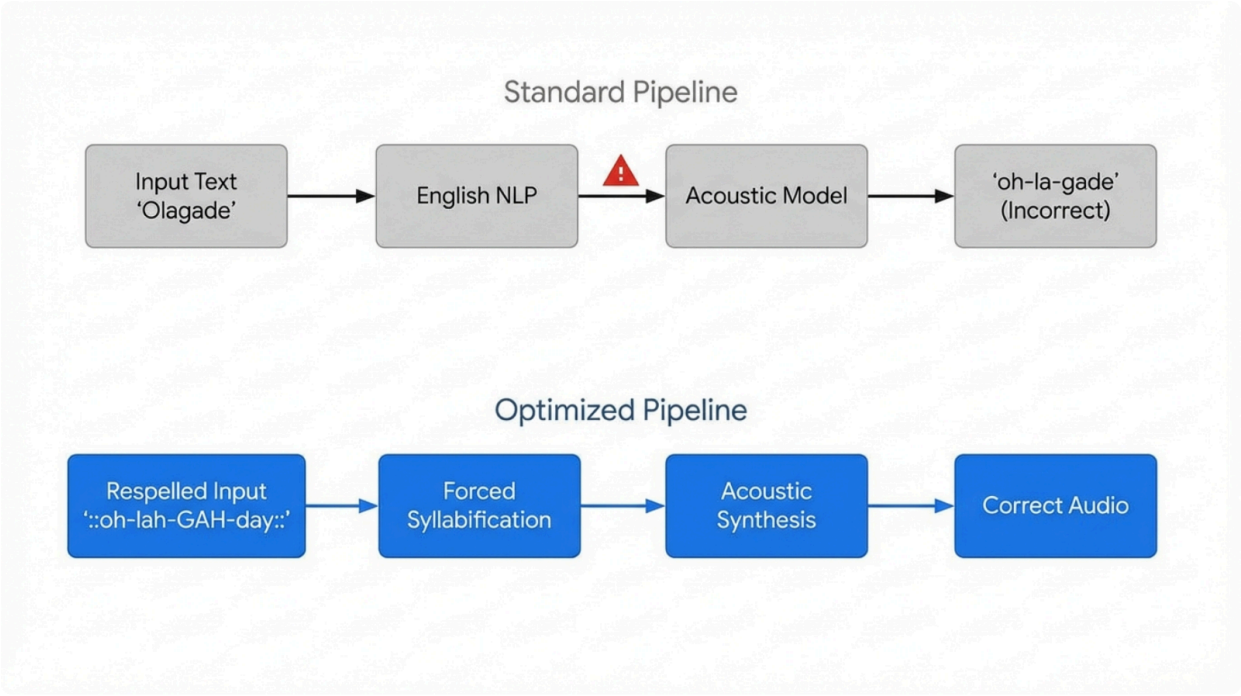
Agglutination and Morphophonemic Sandhi

Kannada operates on an agglutinative framework, chaining discrete, unchanging suffixes to a root word to convey complex grammatical relationships encompassing tense, case, and number. For instance, the root word *Mane* (house) combined with the locative suffix *-alli* (in) creates the compound *Maneyalli*.[5] During this chaining process, the language employs euphonic mutation rules known as *Sandhi* to resolve vowel clashes.[5]

Agama Sandhi specifically dictates the insertion of glide consonants, such as an internal *-y-* or *-v-*, between vowels to prevent a phonetic hiatus.[5] When transliterated words visually expand due to agglutination, standard AI models frequently misinterpret the syllable boundaries. The TTS engine attempts to read the elongated word rapidly without the necessary rhythmic pauses, resulting in an unintelligible blur of syllables. Phonetic respelling methodologies must manually enforce these boundaries using hyphens to force the AI to

process the complex morphology piecemeal.[12]

Bypassing English G2P Models with Phonetic Respelling



Standard English text-to-speech models apply West Germanic phonetic rules to transliterated Kannada, resulting in misassigned stress and alveolar substitution for retroflex sounds. The phonetic respelling protocol bypasses standard G2P processing by manually isolating syllables and enforcing emphasis through capitalization.

The Phonetic Respelling Architecture for AI Override

To compel an American English TTS model to accurately vocalize Kannada, a highly specific respelling architecture must be deployed within the source documents feeding the AI. This system relies on manipulating familiar English orthographic patterns to trigger the desired phonetic output from the neural network's pre-existing phonetic library.[12]

Advanced AI TTS systems interpret specific textual formatting as pronunciation overrides. The rules governing the respelling tables provided in this report are designed to exploit these formatting triggers.[12] Syllabification via hyphenation is the first critical step; words must be broken down into distinct syllables separated by hyphens. Transforming *Ondu* into *ohn-doo* prevents the AI from attempting to blend incompatible consonant clusters.[19]

Furthermore, stress enforcement is achieved via capitalization. Because Kannada is a

syllable-timed language and American English is stress-timed, the AI will naturally misplace emphasis if left unguided. To command the AI to place primary emphasis on a specific syllable, that syllable must be fully capitalized. For example, transforming the word *Mundhe* into *MOON-day* forces the AI to strike the initial syllable with the appropriate acoustic weight.[12]

The weighting of individual vowels requires a standardized mapping protocol to prevent the AI from defaulting to erratic vowel shifts. The short 'a' is mapped to 'ah' or 'uh', while the elongated 'aa' is capitalized as 'AH'. The short 'i' is represented as 'ih' or 'ee', and its elongated counterpart 'ī' is written as 'EE'. The short 'u' is mapped to 'oo', the 'e' becomes 'eh', and the long 'ē' is transcribed as 'ay'. Finally, the standard 'o' is rendered as 'oh'. In certain AI ecosystems, wrapping the entire phonetic spelling in double colons (e.g., ::nah-MUSS-kah-rah::) acts as an explicit algorithmic flag, instructing the processing engine to bypass standard dictionary lookups and read the token strictly phonetically.[12]

Exhaustive Phonetic Lexicon for Spoken Bengaluru Kannada

The following sections provide a rigorously mapped, exhaustive phonetic dictionary. Every Kannada term extracted from the provided research corpus, targeted specifically for migrant integration in Bengaluru, has been transcribed into an AI-optimized phonetic format. This comprehensive map guarantees that the NotebookLM audio generation will articulate the target vocabulary with maximum fidelity to the colloquial dialect, capturing the nuances of social honorifics, transactional logistics, and domestic administration.[5]

Tier 1: Logistical Navigation and Social Hierarchy

The sociolinguistic hierarchy of Kannada strictly divides subjects into the *Mahat* (Rational/Human) and *Amahat* (Irrational/Non-Human) paradigms.[5] Correct pronunciation of these foundational pronouns is vital for the AI hosts to demonstrate cultural respect and accurately teach the language's social dynamics.

English Concept	Kannada Transliteration	AI Phonetic Respelling	Morphosyntactic Context & Usage Notes
I / Me	Naanu	**NAH-noo**	First-person singular. The primary anchor for self-identification in the nominative

			case.
We (Inclusive)	Naavu	**NAH-voo**	First-person plural pronoun. Used inclusively to refer to groups.
You (Respectful)	Neevu	**NEE-voo**	Polite second-person pronoun. The absolute default for strangers, vendors, and colleagues in Bengaluru.
You (Informal)	Neenu	**NEE-noo**	Casual second-person pronoun. Strictly reserved for close intimates or young children; highly offensive if used with service personnel.
He / She / They (Respectful)	Avaru	**AH-vah-roo**	Third-person respectful pronoun. Functions across both genders when respect is required and serves as the general plural.
This (Proximity)	Idu	**IH-doo**	Demonstrative pronoun referring to inanimate objects or animals nearby (Amahat category).

That (Distance)	Adu	**AH-doo**	Demonstrative pronoun referring to inanimate objects or animals located at a physical distance.

Kannada relies on highly efficient spatial and conceptual interrogatives. The AI must appropriately stress the primary syllables to sound inquisitive rather than declarative.[5] Furthermore, standard statements are converted into binary questions via specific suffixes.

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**
What?	Yenu? / Enu?	**YAY-noo**	Universal interrogative for objects and concepts. Can also be utilized to express surprise.
Where?	Elli?	**YELL-lee**	Primary spatial interrogative. Often combined with directional suffixes for navigation.
How much?	Yeshtu? / Eshtu?	**YESH-too**	Critical vocabulary for all financial transactions, market inquiries, and auto-rickshaw negotiations.
Who?	Yaaru?	**YAH-roo**	The specific interrogative used for identifying individuals.

Why?	Yaake?	**YAA-kay**	Interrogative deployed to demand reasoning or understand causation.
Yes/No Suffix	-aa / -vaa	**-AH**	An agglutinative suffix that elongates the terminal vowel of a focal word, transforming a declarative statement into a binary question (e.g., *pustakavaa -> poos-tah-kah-VAH*) .

When navigating the logistical realities of Bengaluru, mastering the polite imperative mood is critical. The AI models must clearly articulate the terminal euphonic vowels (typically ending in -i) that transform harsh root commands into culturally appropriate, respectful requests.[5]

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**
Please come	Banni	**BUN-nee**	The respectful imperative of the root *Baa*. This is the required formal application for daily use.
Please go	Hogi	**HOH-ghee**	The respectful imperative of the root *Hogu*. Essential for directing transport and establishing

			boundaries.
Please give	Kodi	**KOH-dee**	The respectful imperative of the root *Kodu*. Used routinely when requesting items in a shop or asking for tickets.
Please take	Thagonli	**thah-GOHN-lee**	The respectful imperative of the root *Thogo*. Used when handing cash or physical items to a vendor.
Please do / make	Maadi	**MAH-dee**	The respectful imperative of the root *Maadu*. This acts as the universal auxiliary verb frequently appended to English nouns to create "Kanglish" phrases.
Stop	Nillisi	**NILL-lee-see**	The standard respectful command for instructing an auto-rickshaw or cab driver to halt the vehicle.
Look / Inspect	Nodi	**NOH-dee**	Respectful imperative utilized to direct someone's visual attention or request that they

			inspect an object.
Tell / Say	Heli	**HAY-lee**	Respectful imperative used to prompt information, such as asking a vendor to state a final price.
Listen / Ask	Keli	**KAY-lee**	Respectful imperative used to gain someone's auditory attention politely before asking a question.
Want / Need	Beku	**BAY-koo**	An essential modal verb indicating desire or requirement; pervasive in all commercial transactions.
Do not want / need	Beda	**BAY-dah**	The negative counterpart to *Beku*. Highly useful for dismissing hawkers, declining offers, or refusing inflated fares.

Clear pronunciation of spatial markers and directional adverbs ensures accurate communication with transport drivers and delivery personnel. These terms map the physical environment for the speaker.[5]

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**

Here	Illi	**ILL-lee**	Spatial marker indicating immediate physical proximity.
There	Alli	**ULL-lee**	Spatial marker denoting a distant physical location.
Straight	Nera	**NAY-rah**	Directional adverb used extensively in road navigation (e.g., *Nera hogi*).
Left	Edagade / Eda	**YED-ah-gah-day**	Directional noun vital for guiding transport. *Eda* means left, and *gade* means side.
Right	Balagade / Bala	**BULL-ah-gah-day**	Directional noun vital for guiding transport. *Bala* means right.
Forward / Ahead	Mundhe	**MOON-day**	Spatial marker often combined with the imperative for stopping (e.g., *Mundhe nillisi* - Stop ahead).
Backward / Behind	Hindhe	**HIN-day**	Spatial marker indicating the rear or a reversing direction.
Inside	Olagade	**oh-lah-GAH-day**	Useful for directing delivery personnel to leave items

			internally or instructing guests to enter.
Outside	Horagade / Aache	**hoh-rah-GAH-day**	Used to instruct drivers, delivery agents, or waiting individuals to remain externally.

Adjectives and adverbs provide the necessary descriptive nuance for evaluating quality, speed, and quantity during daily interactions. The phonetic weighting ensures these descriptors carry the correct emotional resonance.[5]

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**
A little / Slightly	Swalpa	**SWUL-pah**	A highly utilized adverb that functions both as a measure of quantity and as a polite conversational softener.
Too much / More	Jaasthi	**JAHS-thee**	An adverb indicating excess, frequently deployed as a bargaining tool in markets.
Fast / Quickly	Bega	**BAY-gah**	Adverb of speed utilized to urge drivers or request expedited commercial service.

Slow / Slowly	Nidhana	**nee-DHAH-nah**	Adverb of speed essential for requesting individuals to speak more slowly for comprehension, or to drive carefully.
Good / Well	Chennagide	**chen-NAH-ghee-day**	Adjective/state indicating satisfactory quality, applicable to food, weather, or general wellbeing.
Bad	Ketta	**KET-tah**	Adjective used to describe poor quality, negative states, or undesirable outcomes.
Correct / Okay	Sari	**SAH-ree**	An agreement particle utilized to confirm instructions, agree to a price point, or acknowledge a statement.
Yes	Haudu	**HOW-doo**	The standard affirmative conversational response.
No / Absent	Illa	**ILL-lah**	A negative response indicating disagreement, physical absence, or general

			non-existence.
Big / Large	Dodda	**DOHD-dah**	Adjective that precedes the noun in syntactic order, mirroring English placement.
Small	Chikka	**CHIK-kah**	Adjective utilized to describe diminished physical size.
Hot	Bisi	**BEE-see**	Adjective referring to temperature; applicable to climate, food, and physical objects.
Cold	Thannige	**thun-NEE-gay**	Adjective referring to temperature; specifically utilized for weather and liquids.
New	Hosa	**HOH-sah**	Adjective indicating novelty regarding places or objects.
Old	Hale	**HAH-lay**	Adjective demonstrating the historical phonological shift from the Proto-Dravidian root *Pazhaya*.

Tier 2: Temporal Mechanics, Numerics, and Advanced Morphosyntax

The numeric system in Kannada relies heavily on compound fusions when advancing past the

number ten. Ensuring the AI accurately voices the *Sandhi* (fusion) points where the base units merge is critical to prevent financial misunderstandings.[5] Time markers similarly dictate the rhythm of daily scheduling.

English Concept	Kannada Transliteration	AI Phonetic Respelling	Morphosyntactic Context & Usage Notes
Now	Eega	**EE-gah**	Temporal adverb indicating the present moment; used to demand immediate action.
Later	Aamele	**AH-may-lay**	Temporal adverb extremely common in scheduling discussions or as a delaying tactic.
Today	Ivathu / Ivattu	**ee-VAH-too**	Temporal noun indicating the current calendar day.
Tomorrow	Naale	**NAH-lay**	Temporal noun indicating the future day; frequently utilized to postpone meetings or vendor visits.
Yesterday	Ninne	**NIN-nay**	Temporal noun indicating the previous calendar day.
Morning	Beligge	**bel-LIG-gay**	Temporal noun indicating the

			period before noon.
Afternoon	Madhyahna	**mud-YAH-nah**	Temporal noun indicating midday.
Evening	Sanje	**SUN-jay**	Temporal noun indicating the period before sunset.
Night	Rathri	**RAH-three**	Temporal noun indicating nighttime.
Time	Samaya / Hothu	**SAH-my-ah**	Noun utilized to ask about overall duration or exact clock time.
One	Ondu	**OHN-doo**	The foundational numeric unit, which also functions syntactically as an indefinite article (a/an).
Two	Eradu	**YAY-rah-doo**	Numeric unit central to basic counting and daily market transactions.
Three	Mooru	**MOO-roo**	Numeric unit for three.
Four	Naalku	**NAHL-koo**	Numeric unit for four.
Five	Aidhu	**EYE-doo**	Numeric unit for

			five.
Six	Aaru	**AH-roo**	Numeric unit for six.
Seven	Yelu	**YAY-loo**	Numeric unit for seven.
Eight	Entu	**YEN-too**	Numeric unit for eight.
Nine	Ombathu	**ohm-BAH-too**	Numeric unit for nine.
Ten	Hattu	**HUT-too**	Base unit for ten. Demonstrates the Proto-Dravidian 'p' to 'h' phonetic shift from the root *Pattu*.
Eleven	Hannondu	**hun-NOHN-doo**	Numeric compound demonstrating the Sandhi fusion of *Hattu* (10) and *Ondu* (1).
Twelve	Hanneradu	**hun-NAY-rah-doo**	Numeric compound for 10 + 2.
Thirteen	Hadimooru	**hah-dee-MOO-roo**	The prefix *Hadi-* acts as the 'teen' equivalent in Kannada numerical grammar.
Fourteen	Hadinaalku	**hah-dee-NAHL-koo**	Numeric compound for 10 + 4.
Fifteen	Hadinaidu	**hah-dee-NYE-doo**	Numeric compound

			for 10 + 5.
Sixteen	Hadinaaru	**hah-dee-NAH-roo**	Numeric compound for 10 + 6.
Seventeen	Hadinelu	**hah-dee-NAY-loo**	Numeric compound for 10 + 7.
Eighteen	Hadinentu	**hah-dee-NEN-too**	Numeric compound for 10 + 8.
Nineteen	Hattombattu	**hut-tohm-BAH-to o**	Numeric compound for 10 + 9.
Twenty	Ippattu	**ip-PUT-too**	The base unit for the twenties. The terminal *-ttu* ending denotes a multiple of ten.
Thirty	Moovattu	**moo-VUT-too**	Numeric multiple of ten, used constantly in retail environments.
Forty	Nalavattu	**nahl-ah-VUT-too**	Numeric multiple of ten.
Fifty	Aivattu	**eye-VUT-too**	Numeric multiple of ten.
One Hundred	Nooru	**NOO-roo**	The base unit for hundreds. Vital for discussing rent and handling larger transactions.
One Thousand	Saavira	**SAH-vee-rah**	The base unit for thousands.

A critical syntactic paradigm shift for speakers of fusional languages is mastering the "Experiencer Dative" pattern. In Dravidian linguistics, biological needs and emotional states do not utilize the nominative "I" (*Naanu*). Instead, the experience is framed dynamically as an event that "happens to" the subject, necessitating the dative form "To Me" (*Nanage*).[5]

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**
To Me	Nanage / Nange	**NAH-nah-gay**	The dative form of the first-person pronoun. Serves as the absolute anchor for experiential verbs.
To You (Respect)	Nimage	**NIM-ah-gay**	The dative form of the respectful second-person pronoun.
Known	Gothu	**GOH-too**	Experiential verb indicating knowledge, requiring the dative subject (e.g., *Nanage gothu* - To me it is known).
Unknown	Gothilla	**goh-THILL-lah**	The negative experiential verb. Highly useful for establishing boundaries regarding local comprehension.
Hunger	Hasivu	**HAH-see-voo**	Noun expressed experientially (*Nanage hasivu*

			agtide).
Thirst	Bayarike	**by-YAH-ree-kay**	Noun expressed experientially (*Nanage bayarike agide*).
Pain	Novu	**NOH-voo**	Noun utilized with the dative subject to express physical distress.
Anger	Kopa	**KOH-pah**	Noun expressed dynamically (*Nanage kopa bandide*).
Fear	Bhaya	**BHAH-yah**	Noun expressed experientially (*Nanage bhaya agide*).
Is / Exists	Ide	**EE-day**	Third-person neuter verb utilized to confirm physical availability or existence.

Moving beyond vocabulary, grammatical structuring requires the use of case declensions (*Vibhaktis*) and conjunctions. These phonetic blocks are the agglutinative engines of Kannada, and the AI must append these smoothly to root nouns without introducing unnatural pauses or breaks.[5]

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**
And	Mattu	**MUT-too**	Conjunction used to link similar items

			in lists or connect parallel ideas.
But	Aadare / Adare	**AH-dah-ray**	Conjunction essential for building complex sentences and introducing contrast.
Or	Athava	**AH-thah-vah**	Conjunction deployed for presenting options or choices in a transaction.
Because	Ekeandre	**YAY-ken-dray**	A compound conjunction combining *Yaake* (why) and *Andare* (if said) to provide reasoning.
Therefore / So	Aadirinda	**AH-dee-rin-dah**	Conjunction utilized to indicate causality and result.
In / On	-alli / -nalli	**-ALL-lee**	The Sapthami Vibhakti (Locative case) suffix. It directly replaces English spatial prepositions.
To	-ge / -ige / -kke	**-GAY**	The Chaturthi Vibhakti (Dative case) suffix. Indicates a target destination or

			recipient.
From / By	-inda / -ninda	**-IN-dah**	The Triteeya Vibhakti (Instrumental case) suffix. Frequently serves as an ablative to indicate origin.
Of	-a / -ina / -na	**-AH**	The Shashti Vibhakti (Genitive case) suffix. Indicates possession or relationship.
Plural (Inanimate)	-galu	**-gah-loo**	The primary agglutinative suffix utilized to indicate plurality for non-human objects.
Plural (Human)	-ru / -aru	**-roo**	The specific plural suffix reserved for the Mahat (Rational) category, denoting multiple people or conferring respect.
Only / Itself	-ee / -ne	**-EE**	An exclusive emphatic suffix attached directly to a noun to replace the English "only."
Also / Even	-uu / -nu	**-OO**	An inclusive emphatic suffix

			attached to a noun to replace the English "also."
I go	Hogthini	**HOG-thee-nee**	Present/Future continuous finite verb, formed by combining the root, tense infix, and pronoun suffix.
I come	Barthini	**BAR-thee-nee**	Present/Future continuous finite verb indicating the intention to arrive.
Finished / Done	Aayithu	**AH-yee-thoo**	Past tense completion marker utilized to confirm a task's conclusion.

Tier 3: Domestic Administration, Culinary Taxonomy, and Market Vocabulary

True integration into the sociolinguistic fabric of Bengaluru requires expanding the lexicon beyond transactional logistics and into highly specific domestic and commercial spheres. While corporate environments may default to English, local wet markets and domestic administration rely heavily on native terminology.[5]

English Concept	Kannada Transliteration	AI Phonetic Respelling	Morphosyntactic Context & Usage Notes
House / Home	Mane	**MAH-nay**	Foundational noun for renting properties and directing delivery logistics.

Rent	Baadige	**BAH-dee-gay**	Essential noun for residential negotiation and lease discussions with landlords.
Room	Kone	**KOH-nay**	Noun utilized in real estate discussions and domestic management.
Water	Neeru	**NEE-roo**	Vital noun for discussing utility availability and dining.
Electricity / Power	Current / Vidyut	**CUR-rent**	While *Vidyut* is the formal term, the English loanword *Current* is universally adapted into Kannada syntax.
Kitchen	Aduge mane	**AH-doo-gay MAH-nay**	Compound noun literally translating to "cooking house."
Sweep the floor	Kasam gudisu	**KAH-sum GOO-dee-soo**	Verb phrase used for domestic help. *Kasam* refers to dust, and *gudisu* is the action of sweeping.
Wipe the floor	Nela oresu	**NAY-lah OH-ray-soo**	Verb phrase. *Nela* signifies the floor, and *oresu* is the action of mopping.

Wash the vessels	Paathre tolee	**PAH-thray TOH-lee**	Verb phrase. *Paathre* refers to culinary vessels, and *tolee* means to wash hard objects.
Wash the clothes	Batte ogey	**BUT-tay OH-gay**	Verb phrase. *Batte* means fabric. *Ogey* is a highly specific verb utilized strictly for laundering textiles.
Throw the trash	Kasa haaki	**KAH-sah HAH-kee**	Verb phrase instructing the disposal of waste. *Haaki* means to put or drop.
Vegetables	Tharakari	**thah-rah-KAH-re e**	Noun. Essential category for local market shopping; pluralized with *-galu*.
Onion	Eerulli	**ee-ROOL-lee**	A staple ingredient; native market sellers rarely utilize the English term.
Garlic	Bellulli	**bell-LOOL-lee**	Compound noun combining *Belli* (white/silver) and *Ulli* (bulb/onion).
Ginger	Shunti	**SHOON-tee**	Essential culinary term.
Chilli / Pepper	Menasinakai	**men-ah-sin-KAH-yee**	*Menasu* refers to pepper, while the

			suffix *-kai* denotes the raw vegetable pod.
Coriander Leaves	Kothambari soppu	**koh-THUM-bah-ree SOP-poo**	*Soppu* denotes any leafy green. Used when requesting fresh herbs.
Curry Leaves	Karibevu	**kah-ree-BAY-voo**	Foundational ingredient in South Indian tempering.
Spinach / Greens	Palak soppu	**PAH-luck SOP-poo**	General term for leafy greens in the market.
Potato	Aalu-gedde	**AH-loo GED-day**	*Gedde* denotes a tuber or root vegetable.
Tomato	Tamatekayi	**tah-MAH-tay-KAH-yee**	Often localized to this pronunciation in rural markets.
Carrot	Gajjari	**GUJ-jah-ree**	Used frequently in vegetable markets.
Cabbage	Ele kosu	**YEL-lay KOH-soo**	Compound noun combining *Ele* (leaf) and *Kosu* (cabbage family).
Cauliflower	Hoo kosu	**HOO KOH-soo**	Compound noun combining *Hoo* (flower) and *Kosu*.
Brinjal / Eggplant	Badanekayi	**bah-dah-nay-KAH-yee**	A very common local vegetable, essential for market

			vocabulary.
Lady's Finger	Bendekayi	**ben-day-KAH-yee**	Noun deployed for okra in common local curries.
Cucumber	Southekayi	**sow-thay-KAH-yee**	Common salad ingredient.
Bottle Gourd	Sorekayi	**soh-ray-KAH-yee**	Common native gourd.
Bitter Gourd	Hagalakayi	**hah-gah-lah-KAH-yee**	Recognized for its distinct flavor profile in local dishes.
Ridge Gourd	Heerekayi	**hee-ray-KAH-yee**	Widely utilized in native gravies and chutneys.
Beans	Huralikayi	**hoo-rah-lee-KAH-yee**	Native term for green beans.
Peas	Batani	**bah-TAH-nee**	Common vegetable term.
Lemon	Nimbehannu	**nim-bay-HUN-noo**	Essential culinary term for citrus.
Fruit	Hannu	**HUN-noo**	Refers to ripe fruits and acts as a suffix for specific fruit names.
Banana	Baalehannu	**BAH-lay-HUN-noo**	Compound noun combining *Baale* (plantain) and *Hannu* (fruit).

Mango	Maavinahannu	**MAH-vin-ah-HUN-noo**	Essential seasonal fruit.
Apple	Sebu	**SAY-boo**	Widely used in local fruit stalls.
Grapes	Draakshi	**DRAHK-shee**	Common fruit term.
Orange	Kittale	**kit-TAH-lay**	Common fruit term.
Raw Rice	Akki	**UK-kee**	Specifically refers to uncooked rice grains, a staple grocery purchase.
Cooked Rice	Anna	**UN-nah**	Distinct from *Akki*; refers strictly to rice that has been boiled and prepared for consumption.
Wheat	Godhi	**GOH-dhee**	Staple grain term.
Milk	Haalu	**HAH-loo**	Illustrates the p -> h shift from the Proto-Dravidian *Paal*.
Curd / Yogurt	Mosaru	**moh-SAH-roo**	A critical component of the South Indian diet and culinary vocabulary.
Butter	Benne	**BEN-nay**	Essential dairy product, famous in local dishes like Benne Dose.

Salt	Uppu	**OOP-poo**	Core culinary term.
Sugar	Sakkare	**SUK-kah-ray**	Core culinary term.
Oil	Enne	**YEN-nay**	Refers to any cooking or application oil.
Egg	Motte	**MOHT-tay**	Common grocery item.
Spices	Masale	**mah-SAH-lay**	General term for spice blends.
Meat (General)	Mamsa	**MUM-sah**	Used broadly in non-vegetarian contexts.
Chicken	Koli mari / Koli	**KOH-lee**	Used frequently in local eateries.

Proper noun mastery must also extend to household items and human anatomy to manage medical emergencies and domestic coordination.[5]

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**
Plate	Tatte	**THUT-tay**	Household and dining term.
Cup / Glass	Lota	**LOH-tah**	Used to request clean glasses in small eateries.
Spoon	Chamacha	**CHUM-chah**	Dining implement.
Knife	Chaaku	**CHAH-koo**	Kitchen implement.

Chair	Kurchi	**KOOR-chee**	Household furniture.
Table	Meju	**MAY-joo**	Household furniture.
Bed / Mattress	Haasige	**HAH-see-gay**	Household furnishing.
Blanket	Kambali	**KUM-bah-lee**	Household furnishing.
Towel	Tuvaalu	**too-VAH-loo**	Bathroom necessity.
Soap	Saaboonu	**sah-BOO-noo**	Bathroom necessity.
Broom	Porake	**POH-rah-kay**	Essential for domestic cleaning instructions.
Bucket	Baadali / Baket	**BAH-ket**	*Baket* is widely accepted as functional Kanglish.
Mirror	Kannadi	**KUN-nah-dee**	Household item.
Clock / Watch	Gadiyaara	**gah-dee-YAH-rah**	Used to refer to physical timepieces.
Head	Tale	**TAH-lay**	Anatomical term. Crucial for describing physical ailments (e.g., headache).
Hand	Kai	**KYE**	Anatomical term.

Leg	Kaalu	**KAH-loo**	Anatomical term.
Stomach	Hotte	**HOHT-tay**	Anatomical term.
Blood	Raktha	**RUK-thah**	Essential emergency term.
Fever	Jvara	**JVAH-rah**	A Tatsama (Sanskrit loanword) expressed dynamically.
Cold (Illness)	Negadi	**NEG-ah-dee**	Expressed dynamically (*Negadi agide*).

Finally, integration into Bengaluru is incomplete without a working knowledge of its dynamic street slang and cultural idioms. The AI must accurately deliver these terms to bridge the gap between formal textbook instruction and colloquial reality.[5]

English Concept	**Kannada Transliteration**	**AI Phonetic Respelling**	**Morphosyntactic Context & Usage Notes**
Trouble / Nuisance	Kirik	**KEE-rik**	Slang noun describing an irritating argument, chaotic situation, or problematic individual.
Useless / Bad	Dabba	**DUB-bah**	Slang noun/adjective. Literally translates to "tin box" but is used metaphorically for bad quality.

Fantastic / Great	Bombaat / Sakkath	**BOM-baht / SUCK-kuth**	Slang adjectives used enthusiastically to describe exceptionally good food, weather, or experiences.
Amazing / Perfect	Chindi	**CHIN-dee**	Slang adjective literally meaning "shredded," utilized colloquially to mean something was executed perfectly.
Brother / Dude	Maga / Macha	**MAH-gah / MAH-chah**	Slang noun. Universal markers of camaraderie among peers.
Boss / Teacher	Guru	**GOO-roo**	Slang noun utilized to address auto drivers, tea stall vendors, or strangers respectfully but casually.
Get lost (Vulgar)	Shaata	**SHAH-tah**	Vulgar slang utilized aggressively to dismiss someone.
Lie	Sullu	**SOOL-loo**	Noun deployed to call out falsehoods.
Make space/Compromise	Swalpa adjust maadi	**SWUL-pah ad-JUST MAH-dee**	Idiomatic phrase. A cultural cornerstone of Bengaluru used to

			request physical space or initiate a negotiation.
Shared Coffee	By-two	**BY-too**	Colloquial culinary term for ordering one filter coffee and splitting it into two glasses.
See you later	Hogi bartini	**HOH-ghee BAR-tee-nee**	Idiomatic farewell. Literally translates to "Having gone, I will come," as stating "I am going" is considered inauspicious.
Food done?	Oota aytha?	**OO-tah EYE-thah?**	Idiomatic greeting serving as a primary social icebreaker after saying hello.
Hey bro, what's up?	Yen guru, yen samachar?	**YAYN GOO-roo, YAYN sah-MAH-char?**	Colloquial greeting. Highly casual, used among peers to establish immediate friendly rapport.

Strategic Prompt Engineering within the NotebookLM Architecture

Possessing an exhaustive phonetic mapping dictionary solves the acoustic processing error at the data level; however, deploying this map within Google's NotebookLM requires strategic prompt engineering. NotebookLM's Audio Overview feature allows users to inject "Custom Instructions" to manipulate the behavior, focus, and tone of the AI podcast hosts.[20]

The "Errata Corrige" Injection Protocol

Standard AI podcast generation pulls text directly from the source document. If the AI hosts mispronounce words sourced directly from uploaded Kannada corpus PDFs, the developer must employ a targeted workaround known as the "Errata Corrige" (Correction of Errors) injection.[22]

To intercept the default grapheme-to-phoneme processing, the user must execute the following pipeline:

1. **Create a dedicated Text Note within NotebookLM** titled "Correct Pronunciation of Source Terms."
2. **Populate the note with direct replacement rules** utilizing the phonetic lexicon generated in the previous section. The formatting should be explicit and directive:
   - Replace [Eerulli] with
   - Replace [Chennagide] with [chen-NAH-ghee-day]
3. **Deploy the Custom Instruction Prompt:** Prior to generating the Audio Overview, the user must access the customization menu and input a strict operational directive to bind the AI to the newly created note:
   - *System Prompt:* "You must strictly replace and pronounce the Kannada terms exactly as indicated in the source note titled 'Correct Pronunciation of Source Terms'. Do not use your default pronunciation for any non-English words. When encountering a transliterated word, prioritize the capitalized syllables for heavy vocal stress." [12]

Scripting Host Dynamics and Pedagogical Personas

An educational podcast targeted at Hindi-speaking migrants learning colloquial Bengaluru Kannada requires a specific instructional tone. If left unprompted, NotebookLM AI hosts tend to default to broad, generalized banter and frequently adopt a moralizing tone (e.g., "Learning a new language is just so inspiring!").[23]

To optimize the pedagogical value of the audio, the custom instructions must restrict these default behaviors and enforce a structured linguistic learning environment:

- **Assign Explicit Personas:** Instruct one AI host to act as the "Local Bengaluru Resident" who introduces the Kannada vocabulary, and direct the other host to act as the "Hindi-speaking Newcomer" who asks for clarifications and repeats the phonetic respellings out loud.[23]
- **Enforce Code-Mixing:** Instruct the hosts to comfortably utilize "Hinglish" and "Kanglish" to reflect the authentic linguistic landscape of Bengaluru's corporate and public sectors.[5]
- **Prohibit Moralizing and Digressions:** Insert strict negative prompts into the custom instructions: "Do not judge, give life lessons, or include moralizing comments. Maintain an objective, highly practical tone focused strictly on logistical survival and phonetic accuracy in the city".[23]

Punctuation-Driven Prosody and Pacing

For the phonetic respellings to be processed effectively by the AI, the script's pacing must be artificially controlled. If constructing a manual script or prompt to feed into the NotebookLM framework, punctuation must be deployed not for grammatical correctness, but for prosodic manipulation.[12]

Using commas (,) forces the TTS engine to insert subtle breathing pauses before complex Kannada terms, preventing the audio from running together. Using ellipses (...) creates a dramatic, instructional space between the English translation and the Kannada equivalent, which is crucial for educational podcasts. For example, formatting the text as: "If you want to say 'stop here', you say... ILL-lee NILL-lee-see," forces the AI to slow down and emphasize the phonetic breakdown.[12]

Optimized NotebookLM Custom Instruction Prompt Architecture

LAYER 1: PERSONA ASSIGNMENT

```
1. Primary Goal: Mimic NotebookLM's Style.
2. Format and Structure:
   - Two-voice dialogue: The text should alternate between two podcast
"voices."
3. Style and Tone:
   - Direct and simple language: Avoid complex vocabulary. The language
should be conversational, as if two people are talking informally and
naturally.
   - Non-moralizing: The podcast should not judge, give life lessons, or
include moralizing comments. The tone should be one of objective analysis.
   - Focus on discussion, not full narration.
```

LAYER 2: PRONUNCIATION OVERRIDE

```
Some terms should be replaced as indicated in the source "ERRATA CORRIGE."
Some terms must be strictly pronounced as indicated in the source "Correct
Pronunciation of Source Terms."
```

LAYER 3: PACING RULES

```
The respelling system gives you precise control over exactly how the AI
pronounces each syllable and which syllables receive emphasis.

Format: ::phonetic-spelling::
All respellings are:
- Contained within double colons :: ::
- Written in lowercase letters.
- EXCEPT for emphasized syllables (which are UPPERCASE)
- Useful for creating natural pauses and rhythm.
```

By deploying this tripartite prompt structure in NotebookLM's customization menu, the user forces the AI hosts to adopt relevant personas, strictly adhere to the phonetic respelling dictionary, and utilize punctuation for pedagogical pacing.

Data sources: Reddit, 7taps, LingQ Forum, Ikangai

Expanding the Acoustic Ecosystem: Native Indic TTS Solutions

While NotebookLM's Audio Overview provides a highly accessible, consumer-friendly tool for generating conversational podcasts, its foundational reliance on American English acoustic models presents a rigid ceiling for phonetic perfection.[25] If the phonetic respelling workarounds detailed above do not yield the necessary acoustic fidelity for complex Dravidian phonemes—particularly the retroflex laterals and deep aspiration—migrating the production pipeline to dedicated, multilingual TTS frameworks may be required.

Recent advancements in regional AI development have produced specialized models trained explicitly on the phonotactics and phonetic rules of Indian languages, effectively bypassing the need for manual English phonetic respelling entirely.[27]

Dedicated architectures like Sarvam AI's Bulbul V3 are developed specifically for the Indian linguistic context. These models handle code-mixed speech natively, understanding the shifting acoustic requirements when a speaker pivots between English, Hindi, and Kannada.[27] They correctly weight retroflex consonants and aspiration without requiring phonetic manipulation.[27] Similarly, the open-source Indic Parler-TTS model officially supports numerous Indic languages, including Kannada. It allows developers to prompt the TTS engine with descriptive captions to natively control prosody, emotion, and accent without relying on "hacks".[28]

Furthermore, commercial voice generation platforms such as VoiceGenie and ElevenLabs are trained heavily on Indian regional dialects and names, actively preventing the phonetic degradation common in Western models.[29] These platforms map Kannada graphemes directly to Dravidian phonemes, eliminating the English acoustic intermediary and providing a pure, native-sounding output for high-fidelity educational content.[30]

Conclusion

The successful integration of Hindi-speaking migrants into the Bengaluru metropolitan ecosystem relies heavily on the acquisition of functional, colloquial Kannada. Generating an educational podcast to facilitate this process requires actively overriding the inherent acoustic biases of Western-trained AI speech models. By systematically breaking down the Kannada vocabulary into its constituent syllables, enforcing stress via capitalization, and artificially isolating complex Sandhi fusions, the comprehensive phonetic respelling framework detailed in this report allows standard AI engines to bypass their grapheme-to-phoneme limitations. When this exhaustive linguistic mapping is combined with strategic prompt injections within NotebookLM's Audio Overview framework—specifically controlling host personas, restricting moralization, and pacing delivery via punctuation—creators can generate

culturally resonant, phonetically accurate, and highly effective language training materials.

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